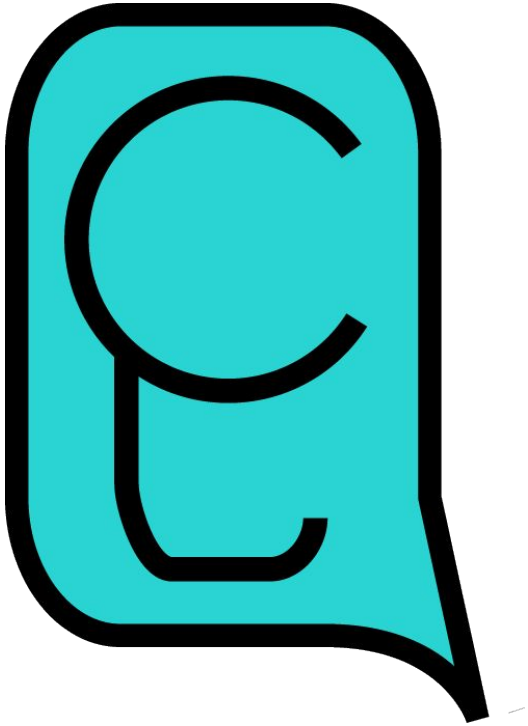


Call Light

Release 1.0.2

Alex Zhu, Arthur Chu, Silas Howe, Griffin Ferguson

The Problem

Patient to nurse communication is inefficient

Pendleton , Bob, et al. “How a Surgical Unit Improved Response to Call Lights.” *How a Surgical Unit Improved Response to Call Lights*, https://accelerate.uofuhealth.utah.edu/connect/how-a-surgical-unit-improved-response-to-call-lights.

Inefficiency

Normal Patient

1. Notice illuminated call light
2. Go to the patient's room
3. Ask the patient what they need
4. Get what they want
5. Bring it to them
6. Leave room

Contagious Patient

1. Notice illuminated call light
2. Go to the patient's room
3. Put on PPE
4. Ask the patient what they want
5. Leave and take off PPE
6. Get what they want
7. Put on PPE
8. Give it to patient
9. Leave room
10. Take of PPE

Normal Patient

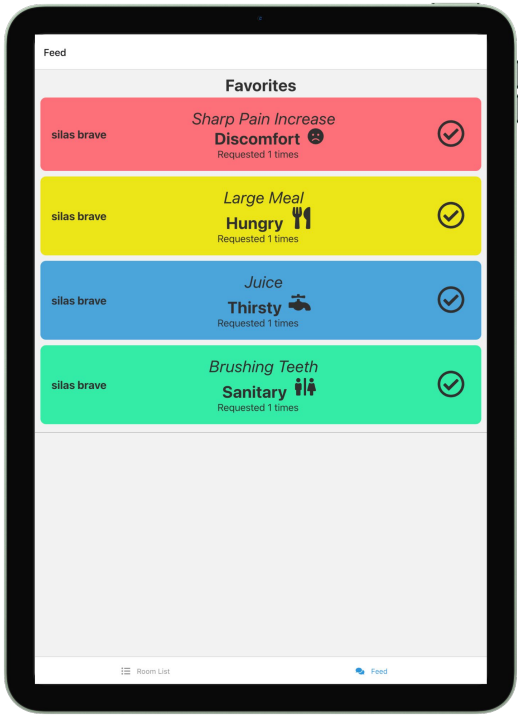
1. Receive notification that patient needs something
2. Get what they want
3. Bring It to them
4. Leave

Contagious Patient

1. Receive notification that patient needs something
2. Get what they want
3. Put on PPE
4. Bring It to them
5. Leave
6. Take Off PPE

Our Solution

With input from HRRMC and Southern New Hampshire Health



Who’s this for?

- Smaller/Lower Budget Hospitals
- Hospital administrators
-

What’s this not for?

- Addressing Emergencies
- Replacing the current system
- Technological integration with other medical hardware
- Deal with patient-nurse relationships

Software Goals

- Keep patient interaction simple
- Maintain privacy
- Relatively easy integration (eg. not using personal devices)

Demo

Username: **room#**
Password: **123456**

(modeling how a nurse would add a new patient to the system)

Existing Features

Interfaces

- Large, readable buttons
- Readable Icons
- Easy to navigate interfaces
- Color Blindness Tolerant
- Device registration

Patient

- Make requests

Nurse

- Mark requests as acknowledged/completed
- “Favorite” a patient
- View requests from all patients
- Easy access to requests from favorites
- See how many times the request has been made

Future Features

In Progress

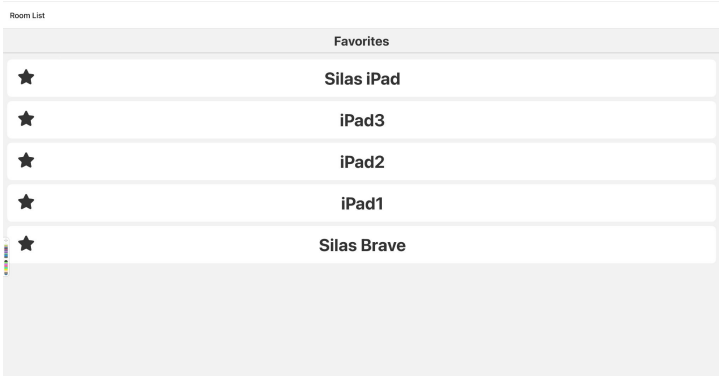
- Patient knows when nurse sees request
- Requests are ranked by time sensitivity
- Quantify Response time

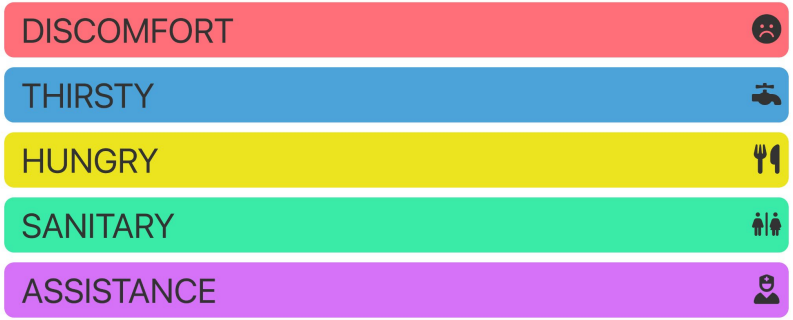
Not Started

- Admin Interface
  - View statistics and track metrics
  - Customize patient choices
- Translation into other languages
- Distinction between nurse and nurse assistant
- Better testing

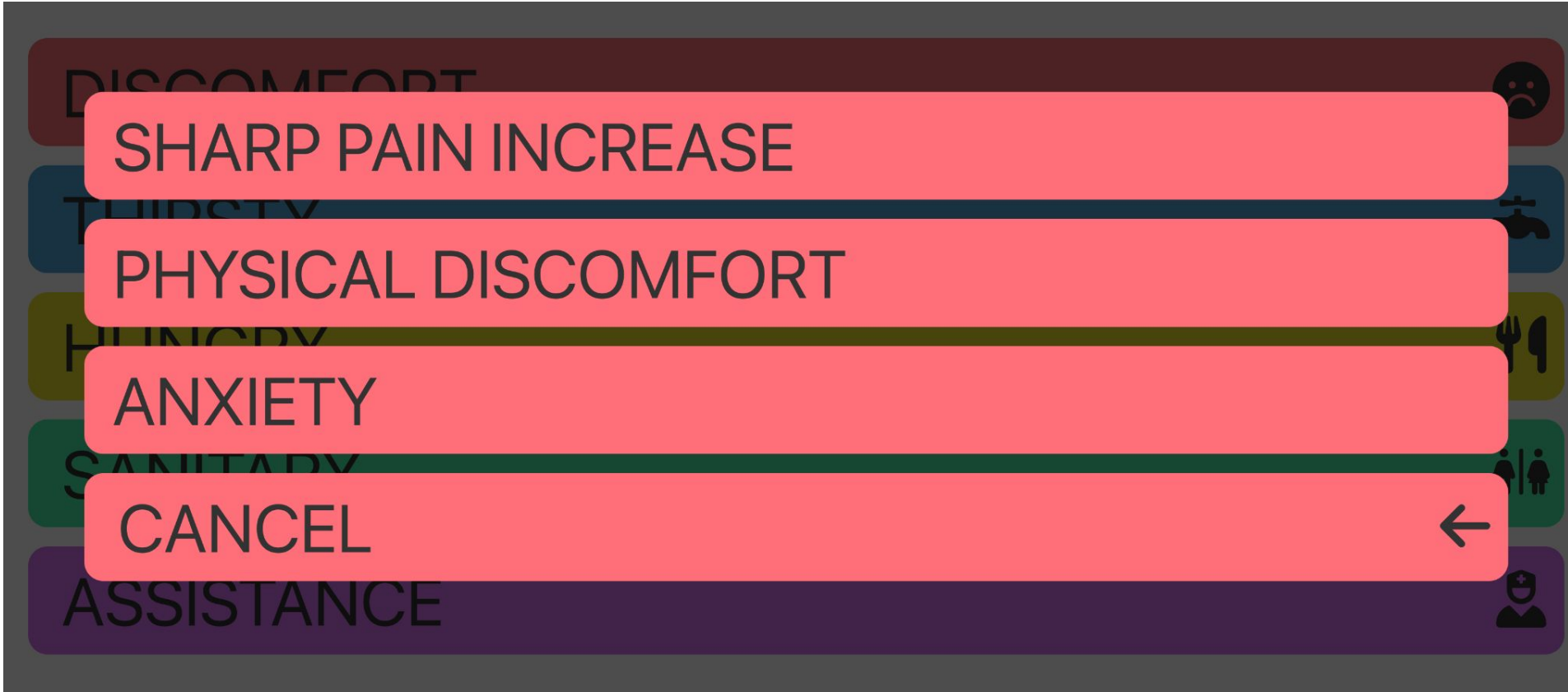
Design Concerns

Usability - Simple Interface





Usability - Large Text



Usability - Color Blindness Tolerant

DISCOMFORT

THIRSTY

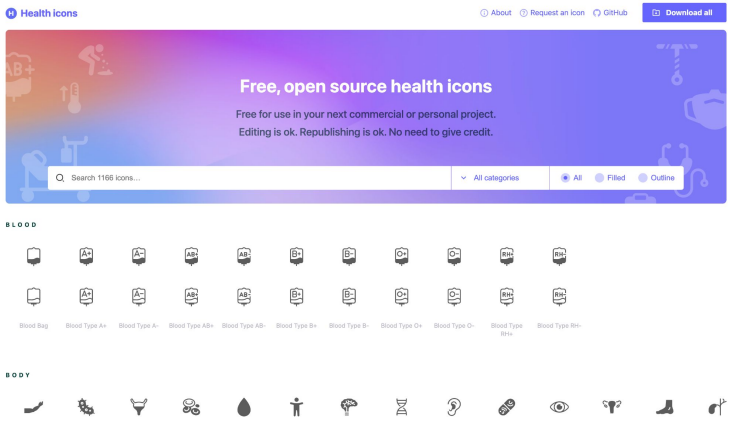
HUNGRY

SANITARY

ASSISTANCE

Usability - Readable Icons

- No international standard
  - Currently using a library from Font Awesome
- healthicons.org is trying change that
- Autism Comm apps



Viability As A Product

- Untapped market (particularly for low budget hospitals)

- Desperate need for increased efficacy for nurse-patient communications
  - Efficiency of caregivers increased
  - Streamlined experience for patients and nurses

- Communicative efficiency → Reduced labor + Better customer experience → Increased profit margins

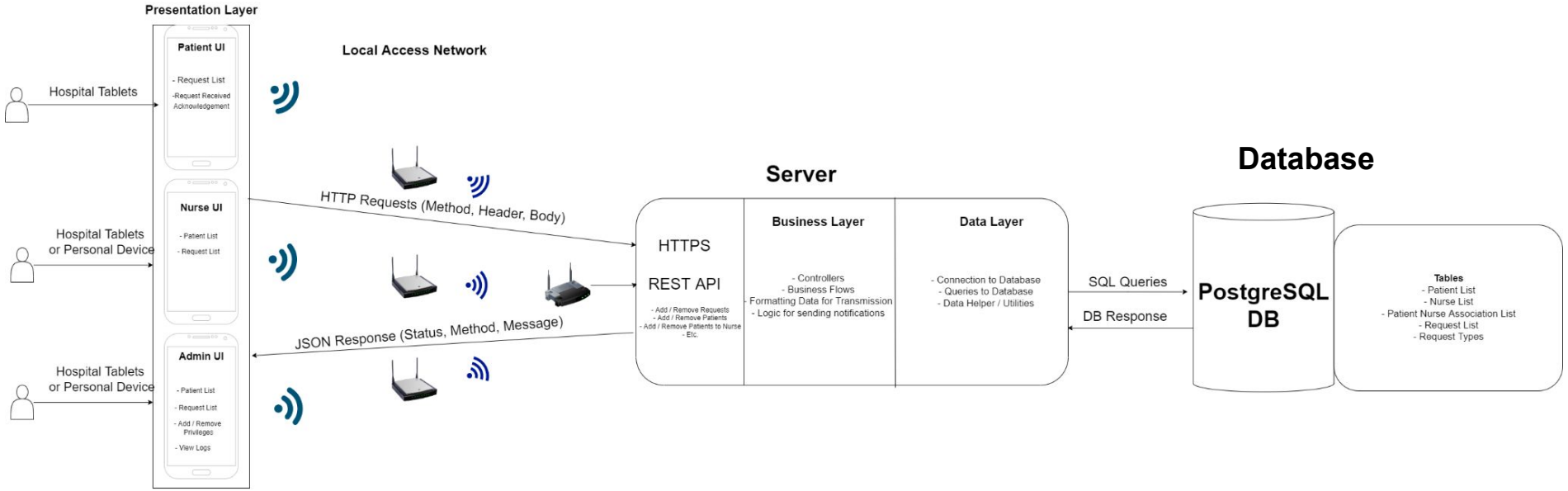
Monetization Strategies

- Proprietary hardware + software [we set it all up]
  - One time payment
  - Subscription
- Paid Software - Unrestricted Hardware
- Free Software
  - Monetization for setup / maintenance

Technical aspects of Call Light

For more detail, refer to the Delivery Documentation

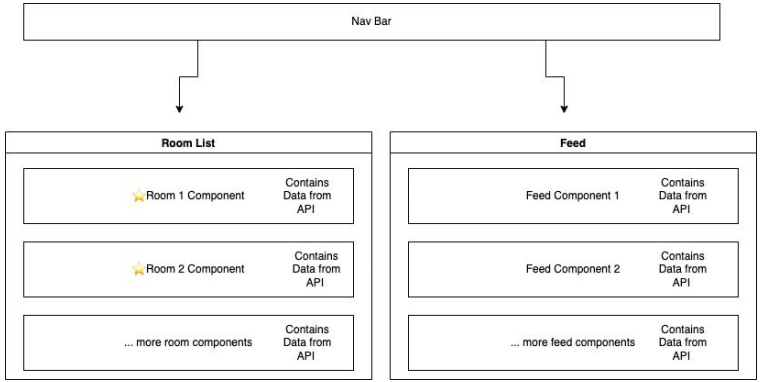
Architecture Specifications



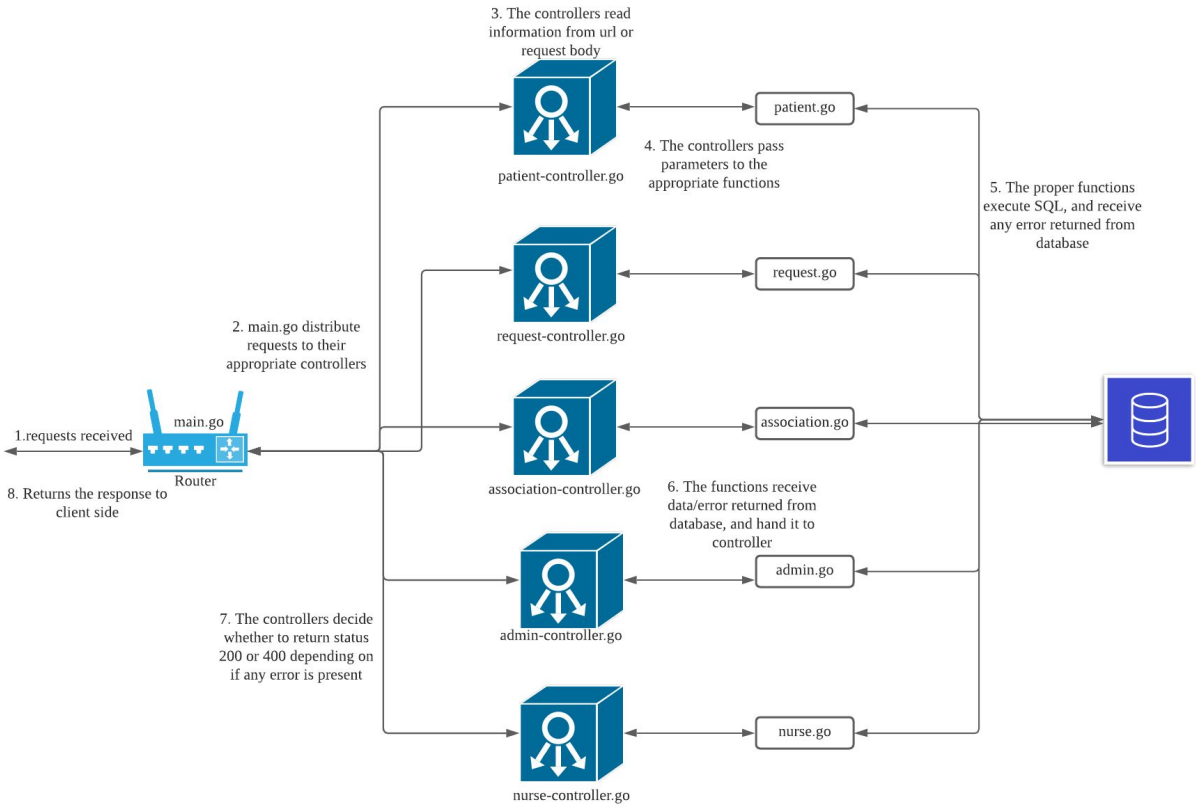
Frontend / UI, Network Configuration, Backend / Server, Database / Persistent Storage

Internal Components of Frontend

- Fetch API to make requests to backend
- Hooks and states to update frontend with request data
- Screens composed of self-made components



Internal Components of Backend



Database Schema

Database Design

Patient

PATIENT_ID	varchar(255) (Primary/Foreign Key)
ROOM	varchar(255)

Request_Type

REQ_TYPE_ID	Int4 (Primary Key)
REQ_TYPE	varchar(255)
LEVEL	int4

Request

REQ_ID	Int4(Primary Key)
PATIENT_ID	Int4 (Foreign Key)
NURSE_ACK	Int4 (Foreign Key)
REQ_TIME	varchar(255)
ACK_TIME	varchar(255)
REQ_TYPE	Int4 (Foreign Key)
IS_FINISHED	bool
FINISH_TIME	varchar(255)

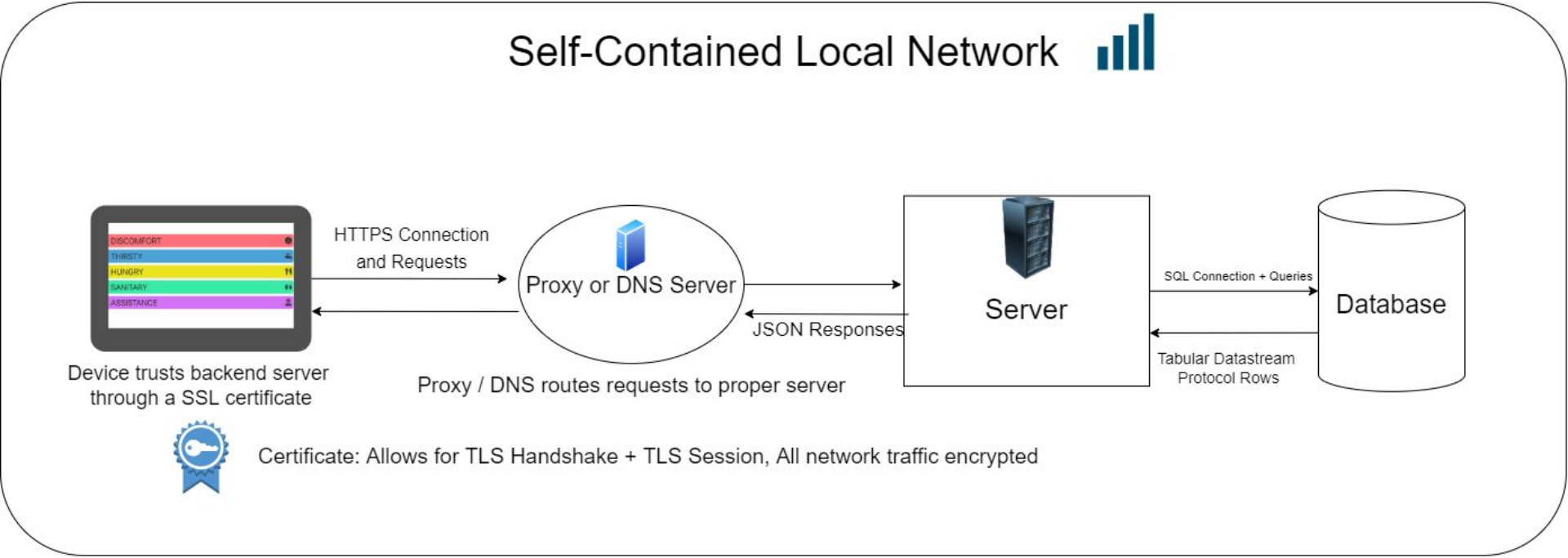
Nurse

NURSE_ID	varchar (Primary Key & Foreign Key)
USERNAME	varchar(255)

Association

ASSC_ID	Int4 (Primary Key)
PATIENT_ID	Int4 (Foreign Key)
NURSE_ID	Int4 (Foreign Key)

Network Flow



Deployment Setup Flow

Frontend:

1. Using Expo, Build into standalone mobile apps

Network Communications:

2. Install certificates into devices (so they trust the server)
3. Setup DNS / Proxy Server on Local Network

Backend:

4. Install server on server machine (including certificate and keys)

Database:

5. Install PostgreSQL
6. Setup tables with proper primary / foreign keys

Validation

- End to end testing of every patient button press combination:
  - Button depth: 49 possible button press combinations

- Minimum of 17 tests per patient button press

Validation

Color and symbol **(3)**

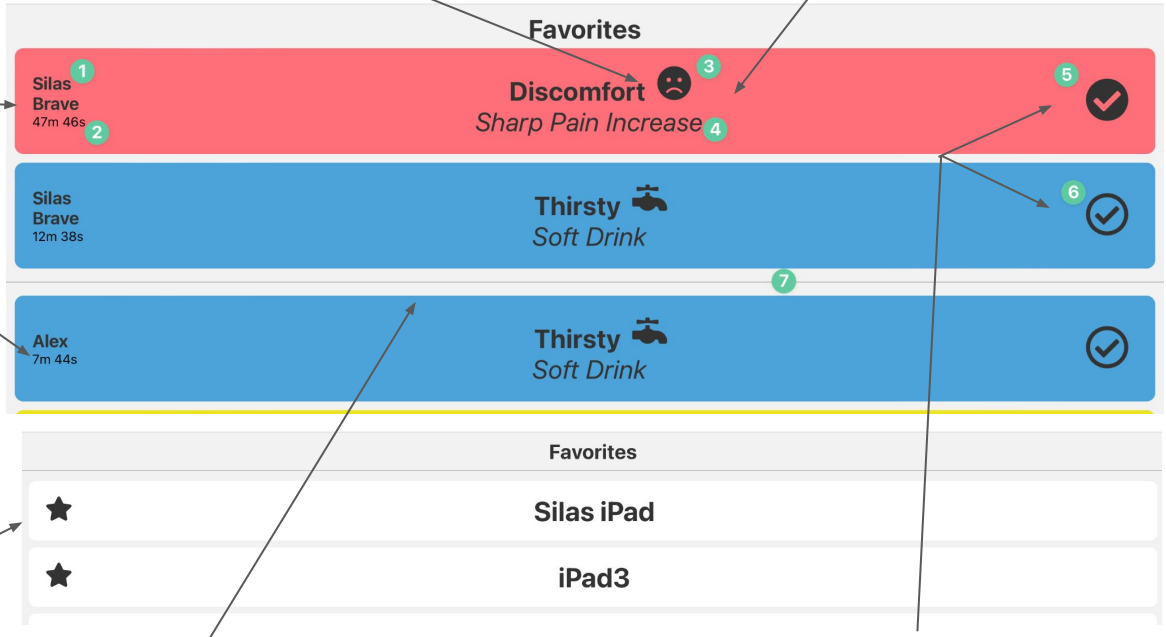
Correct request **(4)**

Population of room name **(1)**

Population of time since request **(2)**

Reverse chronological ordering

Population of patients in patient pool and ability to favorite rooms



Reorganization based on nurse favorites **(7)**

Acknowledge and fulfilled button responses **(5 & 6)**

Validation

- End to end manual testing for all functionality of nurse side functionality
- Manual testing of certificate authentication on multiple devices
- Backend testing
  - Unit testing for handlers/various endpoints

- Missing: minimal non dev team and clinical testing

Ethics Consideration and Software Abuse

- Current build is not requiring patient or nurse names

- False categorizations of requests as complete

- Built in spam control

Teamwork Reflection

- Carefully defined our objectives and limitations in beginning
- Very clear work assignment (frontend & backend groups)
- Targeted primary goals very well
- Integration and teamwork was strong

Challenges/ What We Learned

- It would have been useful to have a longer period for testing/ research
- Having organized platforms for communication (notion & slack) helped greatly
- Better code comments, user flow definition and use of Github issue tracker

Questions?